

Case Study Question Set: Smart City Nepal

Scenario:

Smart City Nepal is a nationwide initiative to digitize urban infrastructure across Kathmandu, Pokhara, and Biratnagar. The platform integrates traffic monitoring, environmental sensors, citizen service requests, and public transport analytics into a unified database ecosystem. It must handle diverse data types, scale across distributed regions, and support real-time decision-making during peak events like Dashain.

Questions:

Enhanced ER & Relational Model

- (a) Draw an EER diagram for Citizens, Vehicles (Public, Private, Emergency), Traffic Signals, and Service Requests.
- (b) Convert the EER into a relational schema and explain indexing strategies for fast lookup of Service Requests by location.

Object & Object-Relational Databases

- (a) Define an object type Sensor with attributes and methods calculateAQI(), detectAnomaly(). Show how inheritance applies for TemperatureCalibratedSensor.
- (b) Explain how PostgreSQL handles UDTs and encapsulated methods compared to a traditional RDBMS.
- (c) Write an OQL query to fetch all sensors that detect anomalies.

Query Processing & Optimization

- (a) Construct a query tree for joining TrafficData with RoadSegments to identify congestion hotspots.
- (b) Apply heuristics to optimize the query and compare two execution plans using cost-based analysis.

Distributed Databases, NoSQL & Big Data

- (a) Explain horizontal and vertical fragmentation strategies for TrafficData and SensorReadings.
- (b) Apply the CAP theorem: During a network partition between Kathmandu and Pokhara, should the system prioritize consistency or availability for live tracking? Justify.
- (c) Design a MapReduce job to calculate average AQI per city. Clearly describe the Map and Reduce phases.

Advanced Models & Applications

- (a) Demonstrate how spatial indexing can be used to locate the nearest hospital during emergencies.
- (b) Explain how temporal databases help track resolution times of citizen service requests.
- (c) Suggest one trigger that could automatically alert authorities when AQI exceeds a threshold.